AI-Powered Collections Strategy

Leveraging Agentic AI for Scalable, Fair, and Effective Debt Management at Geldium

Tata iQ Forage Simulation | Predictive Analytics Team | October 2026

How the System Works

1 · INPUTS

- Customer data: Income, Credit_Score, Credit_Utilization, Debt-to-Income, Loan_Balance, Missed_Payments (12 mo), 6-month payment history, Employment_Status, Tenure
- Data-quality gate first: label clean-up, median imputation + missingness flags (Task 1 cleaning log)

2 · DECISION LOGIC

- Rules first: transparent DTI-band risk tiers (low / medium / high)
- Predictive model only after the go/no-go gate (cross-validated PR-AUC must beat the 0.16 no-skill floor)
- Calibrated probability -> risk score + reason codes

3 · ACTIONS

- Low risk: automated reminder e-mails / SMS
- Medium risk: proactive hardship or limit-review offer
- High risk or vulnerable: human collections review

4 · LEARNING LOOP

- Outcomes tracked: cure rate, roll-rate, complaints, cost per account
- Monthly fairness + drift (PSI) audits
- Rules and models refreshed; kill-switch to manual review

Baseline: 500 customers, 16.0% delinquent - every pipeline step is documented in the Task 1 cleaning log and Task 2 model card.

Role of Agentic AI

AUTONOMOUS (AI acts, no approval needed)	HUMAN OVERSIGHT (approval required)
Nightly segmentation and risk scoring (rules + validated model)	Model go/no-go approval and threshold setting (Task 2 gate)
Reminder e-mails / SMS to low-risk accounts	Hardship and restructuring offers (affordability judgement)
Case routing and priority queues for collectors	Any adverse action: limit cuts, escalation, legal action
Drift / anomaly detection and automatic alerting	Fairness and bias review before any scale-up
Outcome dashboards and standard reporting	Vulnerable-customer exceptions and complaint handling

Principle: AI acts autonomously only where actions are reversible, low-impact and fully explainable; everything affecting a customer's credit terms stays human.

Responsible AI Guardrails

Fairness by design - subgroup audits (Age band, Employment_Status, Location) for equal-opportunity and calibration parity; Location excluded from scoring because it proxies ethnicity and socio-economic status.

Explainability - transparent DTI-band rules today; per-decision reason codes for any future model; predictions communicated as probabilities with uncertainty (executed baseline: no better than chance).

Human-in-the-loop and compliance - no fully automated adverse decisions; adverse-action reason notices; income and debt data minimised to need-to-know; documented model card and full audit trail.

Continuous monitoring - monthly drift (PSI), calibration and complaint-rate reviews; kill-switch back to manual review; fairness re-audit before any scale-up.

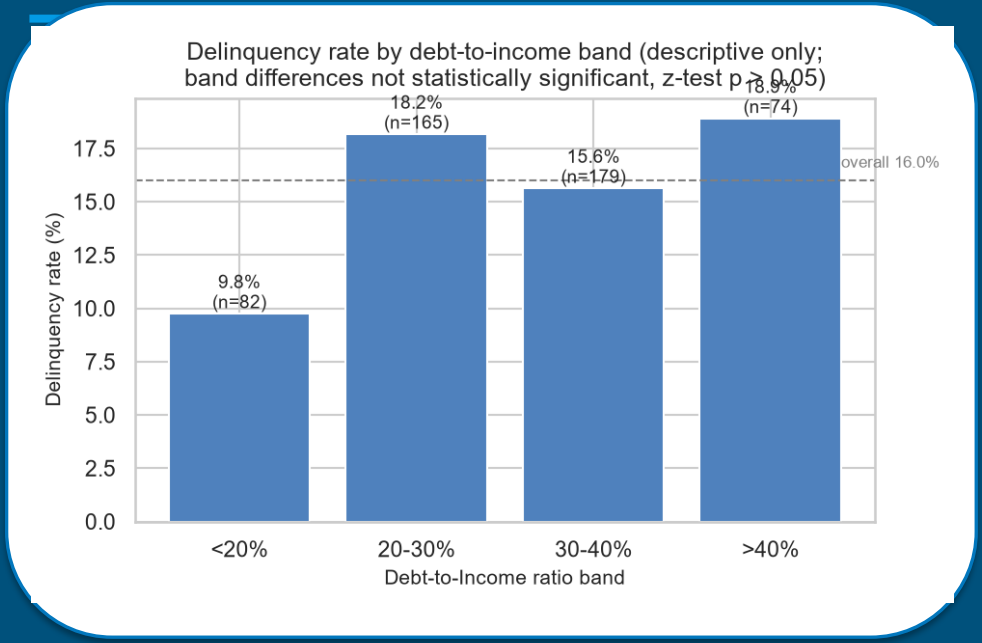
Guardrails are build requirements, not afterthoughts - each maps to a Task 2 evaluation metric and a named owner.

Expected Business Impact

BUSINESS KPIs	CUSTOMER OUTCOMES
15%+ relative delinquency reduction in the pilot cohort (DTI > 40%, 74 accounts) by day 90	Supportive offers before punitive action; hardship routing for vulnerable customers
Lower cost per account: automated reminders replace manual dialling	Fewer intrusive contacts for low-risk customers
Higher cure rate and lower 90+ DPD roll-rate in the targeted band	Transparent reason codes (e.g., 'your debts are close to your income')
Collections capacity freed for complex, high-value cases	Fair outcomes audited by age, employment group and region
Near-zero marginal cost as the portfolio scales	Consistent, documented process customers can question and challenge

Impact targets come from the Task 3 SMART pilot - measured against a matched control, with a scale-or-stop decision at day 91.

Evidence Base: Why Pilot-First



Base rate: 16.0% delinquent (80 of 500 customers)

Strongest observed gradient: DTI > 40% delinquent at 18.9% vs 9.8% below 20% (~2x)

Executed baseline: logistic regression, boosting and dummy all at PR-AUC 0.15-0.17, i.e. the 0.16 no-skill floor

Implication: no model deployment yet - validate the DTI gradient with a controlled 90-day pilot, then scale what works

Definitions: cross-checked against the official Dataset Description Guide (Missed_Payments = 12-month window; Month_1-6 = 6-month window)

Caveat: segments are descriptive, not statistically validated - the pilot exists to validate before scaling (Task 2 go/no-go gate).